

Homework 16

1. The 2004 General Social Survey (GSS) asked respondents whether they had donated blood at least once in the past year. Of the 1271 male respondents, 239 had given blood in the previous 12 months. Of the 1427 female respondents, 201 had given blood. We are interested in drawing conclusions about all adult US residents (in 2004, at least) based on these survey results.

- a. What is our population of interest? (Be careful here! It's not just the people who gave blood.)

All adult residents in 2004

- b. There are two variables of interest in this study. Identify them and classify them (as categorical or numerical).

Variable 1: Gender (categorical) Type of variable
(categorical/numerical):

Variable 2: if donated blood in the past year (categorical) Type of
variable (categorical/numerical):

- c. Make a table showing a few rows of what you imagine the full data set might look like.

Respondent	Gender	Donated blood?
1	Male	Yes
2	Female	Yes
3	Male	No

- d. Calculate the proportions of male ($\hat{p}_{\text{male}}$) and female ($\hat{p}_{\text{female}}$) respondents in the survey who had given blood in the 12 months prior to the survey. Calculate also the difference between the male and female proportions in the order shown below.

$$\hat{p}_{\text{male}} = 239/1271 = 0.188$$

$$\hat{p}_{\text{female}} = 201/1427 = 0.141$$

$$\hat{p}_{\text{male}} - \hat{p}_{\text{female}} = 0.188 - 0.141 = 0.047$$

- e. Calculate the estimated standard errors for the male and female sample proportions based on the sample. Show some of your work.

Males:

$$SE_{\hat{p}_{\text{male}}} = \sqrt{0.188(1-0.188)/1271} = 0.0106$$

Females:

$$SE_{\hat{p}_{\text{female}}} = \sqrt{0.141(1-0.141)/1427} = 0.0092$$

We are interested in the parameter $p_{\text{male}} - p_{\text{female}}$, which represents how much larger the proportion of men who donated blood was than the proportion of women who donated blood. The sample gives us a point estimate ($\hat{p}_{\text{male}} - \hat{p}_{\text{female}}$) for this parameter. As usual, we will also use the data from the sample to judge the precision of this estimate by creating confidence intervals.

- f. Use the new standard error formula (“Pythagorean Theorem”) for a difference in proportions to calculate the estimated standard error for the difference in sample proportions.

$$\sqrt{(0.0106)^2 + (0.0092)^2} = 0.014$$

- g. Use your standard error and the usual $2 \times SE$ rule to calculate a 95% confidence interval for

$$p_{\text{male}} - p_{\text{female}} \pm 2 \times SE_{\text{diff}} = 0.047 \pm 0.028$$

95% CI = (0.019, 0.075) * note this is a rounded CI as it is slightly different from the actual R output that can be found here (0.0184, 0.0760) note sure if this is acceptable or if I should change it

- h. Check your confidence interval using R. You can use the following command:

```
prop.test(x = c(239, 201), n = c(1271, 1427), conf.level = 0.95).
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If the result from the computer tool does not match your own calculations with the standard error, figure out what went wrong either with your use of the computer tool or with your standard-error calculations.

- i. Write a sentence interpreting your 95% confidence interval by filling in the bracketed portions of the following sentence:

I am 95% confident that the proportion of males in the US adult population in 2004 who gave blood is between 1.9% and **7.5%** higher than the proportion of females in **the US adult population in 2004** who gave blood.